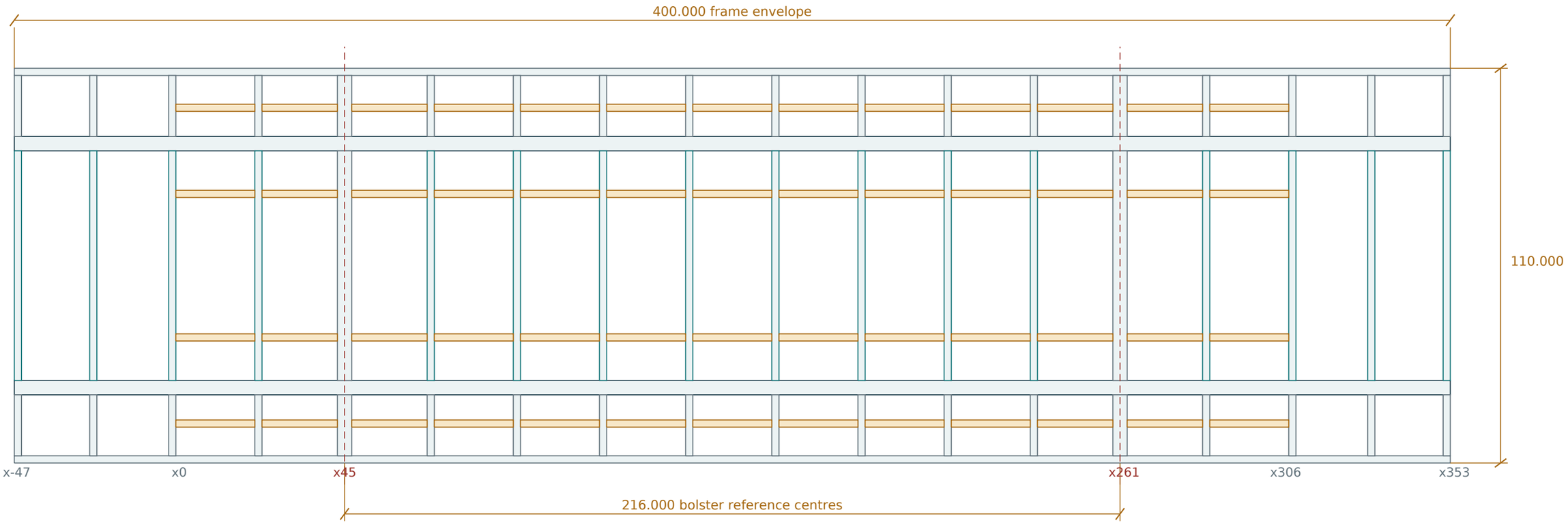
INTEGRATED UNDERFRAME / STUDY CONFIGURATION J1a

110 reference member IDs; 16 I-CM6 + 52 I-S32 section changes. Baseline bolster geometry retained; I-BA and I-BT are NOT selected.

J1.1 FRAME PLAN / x-y / INCHES



Cabin inside end walls x0 / x306. Main centre lines y +/-34. Seat-rail centre lines y +/-20 / +/-44.

Teal = I-CM6; gold = I-S32; gray = retained members. Drawn abutting ends are not released welded connections.

J1.2 MEMBER SCHEDULE / NOMINAL SECTION AND END-PLANE LENGTH

Group	Qty	Study section	Length / in
Main rails	2	HSS 12 x 4 x 1/4	400.000
Central ordinary spans	16	HSS 6 x 2 x 3/16	64.000
Seat support rails	36 / 16	HSS 3 x 2 x 3/16	22.000 / 21.000
Deep central bolsters	2	HSS 12 x 4 x 1/4	64.000

J1.3 COORDINATED LIMITS

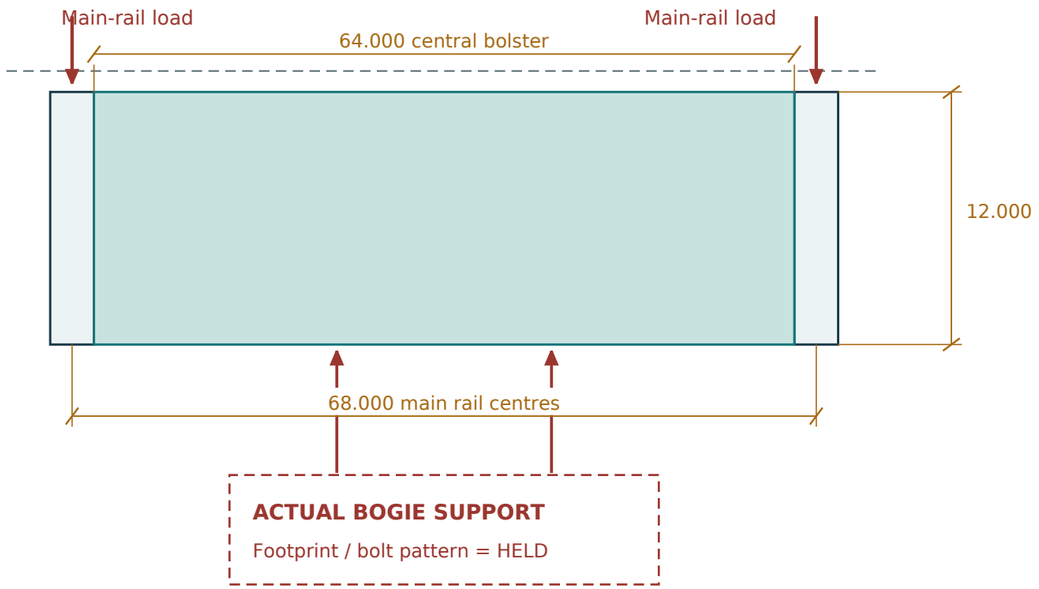
Top of all steel: z31.003937 in. Finished floor study: z32.000 in. Main and central bolster depth: 12.000 in. No bogie mounting holes, reinforcing diaphragm or production weld is assigned.

Actual cuts require end preparation, fit-up allowance, distortion control and engineer release.

MAIN RAILS / BOLSTER / BOGIE INTERFACE

Load-path study only. Baseline HSS centre bolster is drawn for continuity; its local support, retention and main-rail joints remain unresolved.

J2.1 TRANSVERSE SUPPORT / x45 OR x261 / NOT TO SCALE



Arrows indicate load direction only. Their spacing is NOT a bearing / side-bearer dimension.

No I-BA angles shown: earlier weld/angle sensitivity does not establish a viable rigid joint.

J2.2 OEM RETURN FIELDS / NEEDED BEFORE SUPPORT DESIGN

Required interface	Actual value / source
Bogie model / drawing rev	_____
Support footprint / contact law	_____
Vertical / lateral / longitudinal limits	_____
Yaw / roll / pitch travel and stops	_____
Side-bearer preload / friction / wear	_____
Retention / uplift / jacking path	_____
Mount holes / fasteners / torque	_____
Brake equipment / hose envelope	_____
Fatigue life / load spectrum basis	_____

J2.3 MAIN-RAIL GLOBAL SCREEN / BALANCED 2g

Case / quantity	Calculated study result
Coach mass / vertical multiplier	13,200 kg / 2.0g study
Each rail / front and rear support	64.724 / 64.724 kN
Max absolute main-rail moment	54.538 kN m
Gross vertical M/S stress	168.087 MPa
Maximum absolute beam deflection	19.969 mm

Constant EI; free 92 in overhangs; distributed / discrete gravity inputs. No deflection limit accepted.

J2.4 NEXT CONNECTION DESIGN INPUTS

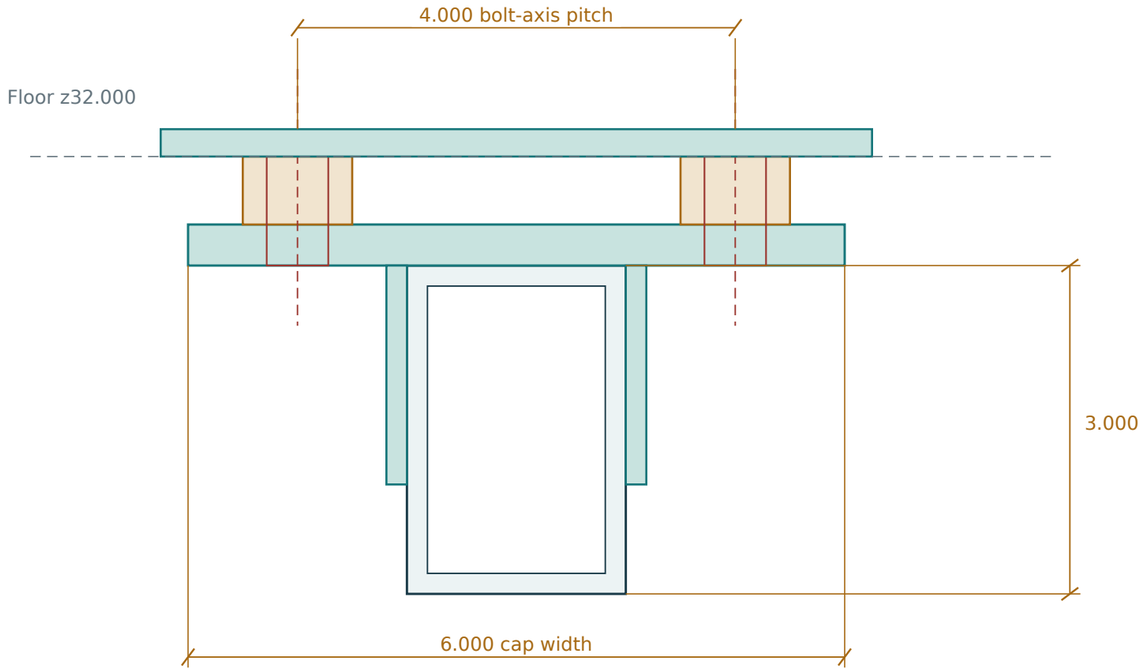
Gravity enters the mains through floor and seat crossmembers. Each main transfers load through both bolster nodes to the bogie support. Braking, lateral, torsional and uplift forces require separate load cases and actual restraint definitions.

Global beam stress does not resolve thin-wall load introduction, weld fatigue, centre-bearing plates or bogie attachment. All remain held.

SEAT / FLOOR / HARDPOINT MOCKUP

Full-size physical mockup proposed; drawings are reduced-scale and not templates. Final I-SH1 geometry retained. No occupied or proof test authorized.

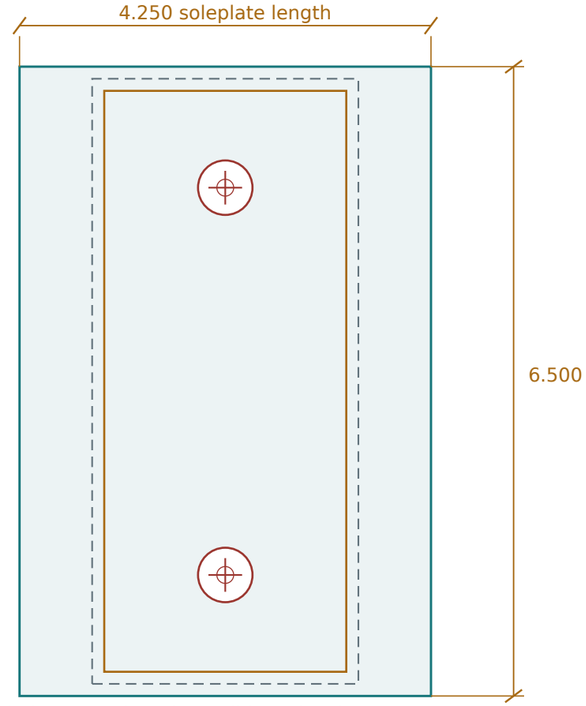
J3.1 ONE FOOT / TRANSVERSE y-z SECTION / DIMENSIONS IN INCHES



Spacer OD 1.000; bore .5625; nominal length 15.775 mm (.621063 in).

Rigid spacer bypasses floor core. Actual clamping and seat-to-sole connection are held.

J3.2 LOCAL PLAN / FOOT CENTRE AT x97, y20



Red = 9/16 study bores / axes; bolt, nut and washer solids omitted.

Dashed opening: 2.750 x 6.250 finished OEM-closed edge; no raw honeycomb cut.

J3.3 ONE-FOOT STUDY PARTS / SIX STEEL PIECES

Component	Qty	Study size
Cap / A36 candidate	1	2.500 x 6.000 x .375
Cheeks / A36 candidate	2	2.500 x 2.000 x .1875
Sole / A36 candidate	1	4.250 x 6.500 x .250
Rigid spacer / steel candidate	2	1.000 OD; .5625 ID; 15.775 mm
Bolt assemblies / Grade 5 candidate	2	1/2-13; length / nut / torque HELD

J3.4 MOCKUP CHECKS / ENGINEER TO RELEASE

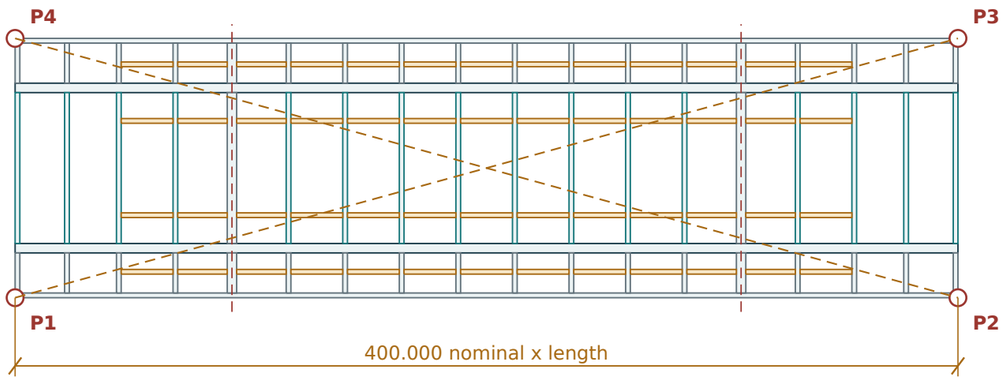
Full bench feet remain at 16 x 24 in centres. Use the actual seat, panel, fasteners and tools. Check seal, edge cover, tool access, spacer contact, weld access and seat height before concealment.

Candidate welds carried from I07; no new weld size, bolt torque, test force or acceptance tolerance is authorized here.

FIRST ARTICLE / DATUMS AND MEASUREMENT RECORD

Build authorization and acceptance limits require responsible engineer release. This sheet defines what to record; blank fields are not zero tolerances.

J4.1 INSPECTION DATUMS / RETAINED NOMINAL GEOMETRY



A: steel-top plane z31.003937. B: vehicle centre plane y0.

C: front cabin inside end-wall station x0. Forward = -x.

J4.2 SETUP RECORD / TRACEABLE FIRST ARTICLE

Record	Actual
Serial / revision / date	_____
Instrument / calibration / method	_____
Temperature / unloaded support	_____
Fixture / shims / restraint map	_____
Material heat / wall / corner profile	_____
WPS / weld map / welder identity	_____
Responsible engineer / inspector	_____

J4.3 MEASUREMENT SCHEDULE / ENTER ACCEPTED LIMIT AND MEASURED VALUE

ID	Feature / method	Nominal study target	Engineer limit	Measured / record
M01	Frame envelope P1-P2 / P1-P4	400.000 / 110.000 in	_____	_____
M02	Main rail y centre lines	-34.000 / +34.000 in	_____	_____
M03	Bolster x reference centres	45.000 / 261.000 in	_____	_____
M04	Top steel plane / floor reference	31.003937 / 32.000 in	_____	_____
M05	Diagonals P1-P3 / P2-P4	414.849370 in each*	_____	_____
M06	Local crossmember / seat rail orientation	6 x 2 / 3 x 2 in	_____	_____
M07	Seat foot / bolt pitch / spacer stack	J03 + actual supplier parts	_____	_____
M08	Actual bogie mount / bearing interface	J02 OEM return required	_____	_____

*Calculated outer-envelope diagonal only. As-built points must reference the same corners; fixture and weld distortion are separately assessed.

Hold: no production repetition until geometry, joint inspections, test results and all deviations have a signed disposition.